Photochemistry (3)

2017

* Deduce an expression for the kinetics of photochemical reaction between H_2 and Br_2.

Photochemical reaction between H_2 and Br_2:

The overall photochemical reaction between H_2 and Br_2 is-

$$H_2 + Br_2 \xrightarrow{h\nu} 2HBr$$

The mechanism that has been proposed for this reaction is as follows-

1. Initiation step:

$$Br_2 + h\nu \xrightarrow{k_1} 2Br \cdot \quad , \text{Rate} = k_1 I_{abs}$$

2. Propogation step:

(i) $Br\cdot + H_2 \xrightarrow{k_2} HBr + H\cdot$, $\text{Rate} = k_2 [Br\cdot][H_2]$

(ii) $H\cdot + Br_2 \xrightarrow{k_3} HBr + Br\cdot$, $\text{Rate} = k_3 [H\cdot][Br_2]$

3. Inhibition step:

$$H\cdot + HBr \xrightarrow{k_4} H_2 + Br\cdot \quad , \text{Rate} = k_4 [H\cdot][HBr]$$

4. Termination step:

$$Br\cdot + Br\cdot \xrightarrow{k_5} Br_2 \quad , \text{Rate} = k_5 [Br\cdot]^2$$

The overall rate of formation of HBr is

$$\frac{d[HBr]}{dt} = k_2 [Br\cdot][H_2] + k_3 [H\cdot][Br_2] - k_4 [H\cdot][HBr] \quad \ldots\ldots (i)$$

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For steady-state condition-

$$\frac{d[H]}{dt} = k_2[Br][H_2] - k_3[H][Br_2] - k_4[H][HBr] = 0 \quad \text{---(ii)}$$

$$\frac{d[Br]}{dt} = k_1 I_{abs} - k_2[Br][H_2] + k_3[H][Br_2] + k_4[H][HBr] - k_5[Br]^2 = 0 \quad \text{---(iii)}$$

Now, adding eqn (ii) and (iii) we get-

$$k_1 I_{abs} - k_5[Br]^2 = 0$$

$$\Rightarrow [Br]^2 = \frac{k_1 I_{abs}}{k_5}$$

$$\therefore [Br] = \left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2}$$

putting the value of [Br] in eqn (ii) we get-

$$k_2 \cdot \left(\frac{k_1 \cdot I_{abs}}{k_5}\right)^{1/2} [H_2] - k_3[H][Br_2] - k_4[H][HBr] = 0$$

$$\Rightarrow [H]\{k_3[Br_2] + k_4[HBr]\} = k_2 \cdot \left(\frac{k_1 \cdot I_{abs}}{k_5}\right)^{1/2} \cdot [H_2]$$

$$\Rightarrow [H] = \frac{k_2\left(\frac{k_1 \cdot I_{abs}}{k_5}\right)^{1/2} \cdot [H_2]}{k_3[Br_2] + k_4[HBr]}$$

Now, putting the value of [H] and [Br] in eqn (i) we get-

$$\frac{d[HBr]}{dt} = k_2 \cdot \left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2} [H_2] + k_3 \times \frac{k_2\left(\frac{k_1 \cdot I_{abs}}{k_5}\right)^{1/2} \cdot [H_2][Br_2]}{k_3[Br_2] + k_4[HBr]} - \frac{k_4 k_2 \left(\frac{k_1 \cdot I_{abs}}{k_5}\right)^{1/2} \cdot [H_2][HBr]}{k_3[Br_2] + k_4[HBr]}$$

$$\frac{d[HBr]}{dt} = k_2[H_2]\left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2}\left\{1 + \frac{k_3[Br_2] - k_4[HBr]}{k_3[Br_2] + k_4[HBr]}\right\}$$

$$\frac{d[HBr]}{dt} = k_2[H_2]\left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2}\left\{\frac{k_3[Br_2] + k_4[HBr] + k_3[Br_2] - k_4[HBr]}{k_3[Br_2] + k_4[HBr]}\right\}$$

$$\frac{d[HBr]}{dt} = \frac{2k_2[H_2]\left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2} \cdot 2k_3[Br_2]}{k_3[Br_2] + k_4[HBr]}$$

$$\frac{d[HBr]}{dt} = \frac{2k_2\left(\frac{k_1 I_{abs}}{k_5}\right)^{1/2}[H_2]}{1 + \frac{k_4[HBr]}{k_3[Br_2]}}$$

This is the expression for the kinetics of photochemical reaction between H_2 and Br_2.

$$\left[\text{Quantum yield} = \frac{\frac{d[HBr]}{dt}}{I_a}\right]$$

***State and explain the laws of photochem
2015

Laws of photochemistry:

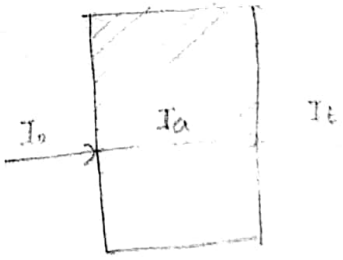
1. Grotthuss - Draper law:

" when light falls on any substance, only the fraction of incident light which is absorbed by the substance can bring about a chemical change: reflected or transmitted light do not produces any such effect."

Explanation:

If I_0 is the intensity of incident light and I_t is the intensity of transmitted light, then the intensity of light absorbed I_a is -

$$I_a = I_0 - I_t \quad \text{---- (i)}$$



From Beer- Lambert law-

$$I_t = I_0 e^{-\varepsilon c l}$$

Therefore eqn (i) becomes-

$$I_a = I_0 - I_0 e^{-\varepsilon c l}$$

$$\Rightarrow I_a = I_0 (1 - e^{-\varepsilon c l})$$

This law is a qualitative treatment ~~of~~ relationship between light absorbed by reactants and occurrance of chemical change.

2. Law of photochemical equivalence:

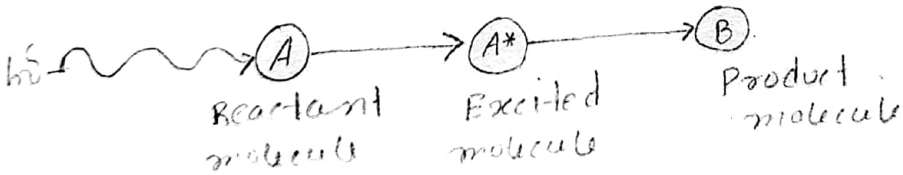
Einstein applied the quantum theory to photochemical reactions which stated as—

"When an atom or molecule absorbs light of a given frequency, it absorbs one quantum only."

The photochemical importance of the above law emphasized by Stark. Einstein considered the work of stark and restated his law as—

"Each molecule which takes part in a chemical reaction absorbs one quantum of light which induces the reaction."

Explanation:



Here molecule A absorbs a photon of radiation and get activated. The activated molecule 'A' decomposes to product B. We can write the above system as—

$A + h\nu \longrightarrow A^*$

$A^* \longrightarrow B$

$A + h\nu \longrightarrow B$ (overall reaction)

One particle is excited for each quantum of radiation absorbed –

$$E = h\nu$$

For 1 mole

$$E = Nh\nu$$

$$\Rightarrow E = \frac{Nhc}{\lambda} \quad [\because \nu = \frac{c}{\lambda}]$$

where,

E = Energy absorbed per mole.

h = Plank's constant

ν = frequency of absorbed radiation

N = Avogadro's number

c = velocity of light

λ = wavelength.

*** Define the term: resonance fluorescence, sensitized fluorescence and quenching fluorescence.

Resonance fluorescence:

In certain cases, the fluorescent light has the same frequency as that of incident light. This fluorescent light is called resonance radiation and the phenomenon is known as resonance fluorescence. This phenomenon is analogous to resonance of sound.

For example- when mercury vapour is exposed to ultraviolet radiations, it absorb the radiation of wavelength 2537 A°. If excited electrons return to the ground state, they emit the radiations of the same wavelength.

$$Hg + h\nu \longrightarrow Hg^*$$
$$Hg^* \longrightarrow Hg + h\nu$$

Sensitized fluorescence:

A substance which is normally non-fluorescent may be made fluorescent in the presence of other fluorescent substances. This phenomenon is known as sensitized fluorescence.

For example- if the vapour of Tl is added to Hg vapour and then exposed to the radiations of wavelength 2537 A°, the vapour of Tl fluoresces.

$$Hg + h\nu \longrightarrow Hg^*$$
$$Hg^* + Tl \longrightarrow Tl^* + Hg$$
$$Tl^* \longrightarrow Tl + h\nu'$$





Quenching fluorescence:

When a photochemically excited atom has a chance to undergo collision with another atom or a molecule, before it fluoresces, the intensity of the fluorescent radiation may be stopped. This phenomenon is known as quenching of fluorescence.

For example – the excited Hg^* atom undergo collision with ~~another~~ H_2 molecule and thus it dissociates.

$$Hg^* + H_2 \longrightarrow Hg + 2H$$

Explain the terms fluorescence and phosphorescence with suitable examples.

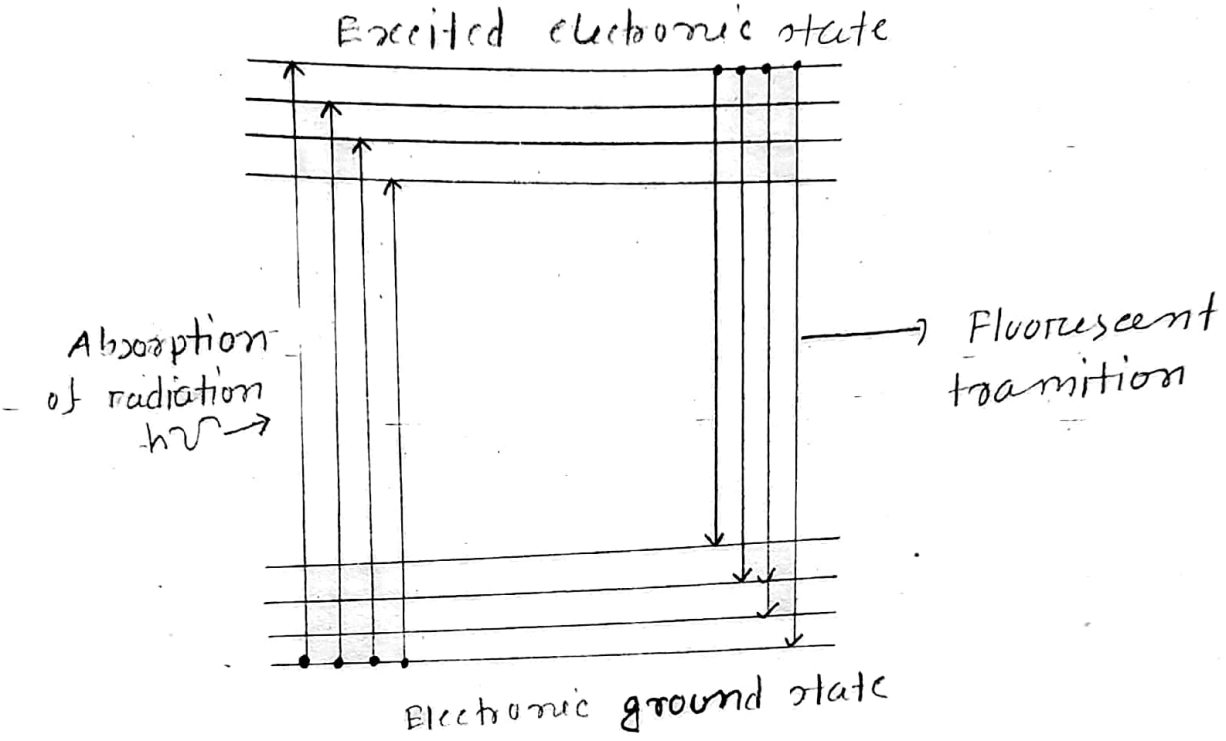
Fluorescence:

When a beam of light is incident on certain substances, they emit visible light or radiations and they stop emitting light or radiation as soon as the incident light is cut off. This phenomenon is known as fluorescence.

Examples:

~~Chloroph~~ chlorophyll present in green leaves show the phenomenon of fluorescence. Acetone absorbs at 2700Å and emits blue fluorescence.

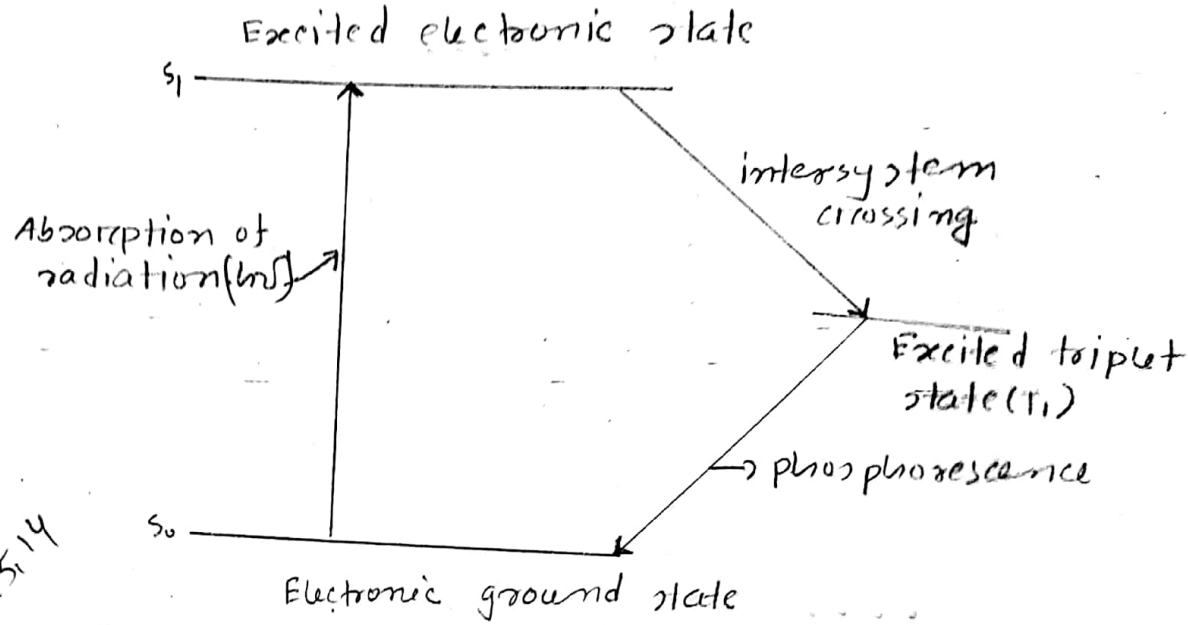
Energy diagram:



Phosphorescence: When a beam of light radiation is incident on certain substances, they emit light continuously even after the incident light is cut off. This type of delayed fluorescence is called phosphorescence.

Examples:
Sulphides of calcium, barium, strontium shows phosphorescence phenomenon.

Energy diagram:



D. 29.5.14

Discuss the application of fluorescence and phosphorescence.

Applications of fluorescence:

1. Fluorescent lightening is more energy efficient than incandescent lighting elements.
2. The determination of uranium in salts by fluorescence.
3. Generally, inorganic ions do not exhibit fluorescence. However some inorganic ions form fluorescent chelate with non-fluorescent organic molecules.

Examples are-

→ Determination of ruthenium ion in the presence of platinum metals.

→ Determination of aluminium in alloys.

→ calcium can be estimated by precipitating it with 2-(2-hydroxyphenyl)-benzoxazole in the presence of tartrate.

(4) In acid-base titrations, fluorescent indicators are used.

(5) Fluorescence-labelled antibodies are used for the detection of antigens.

(6) Determination of vitamin B_1 and B_2 are done by fluorescent method.

7. The phenomenon of fluorescence is being utilised in examining condition of food stuffs.

8. The phenomenon of fluorescene has been used in the analysis for drugs and dyes, textile and paper industry etc.

Application of phosphorescence:

1. Determination of aspirin in blood serum with high sensitivity by phosphorimetry.
2. Low concentrations of procaine, cocaine, phenobarbital and chloropramazine in blood serum have been determined by phosphorimetry with extraction procedures.
3. Phosphorimetry has been employed in combination with thin layer or paper chromatography.
4. Phosphorescence is also used to analyze water, air and chemical pollutions.
5. Phosphorescence light is used on glowing suits by dancers.

** Short note 2

* Define photosensitized reaction and give examples using different types of photosensitizer.

Photosensitized reaction:

In some photochemical reaction, the reacting species itself are unable to absorb radiation directly but these reaction ~~are~~ can be made sensitive by adding a small amount of foreign material which can absorb light and stimulate the reaction without itself taking part in the reaction. These reaction are called photosensitized reaction and the phenomenon is known as photosensitization.

Examples:

a) Reaction sensitized by mercury atoms:
The reaction between hydrogen and oxygen in the presence of mercury vapour under the influence of light radiation —

$$3H_2 + 2O_2 \xrightarrow[\text{light}]{Hg} 2H_2O + H_2O_2$$

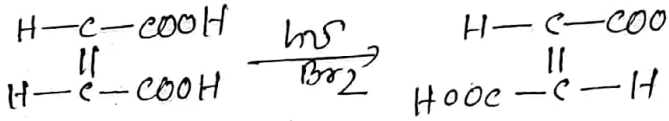
b) Chlorine as a photosensitizer:

Decomposition of ozone in the presence of chlorine –

$$2O_3 \xrightarrow[h\nu]{Cl_2} 3O_2.$$

c) Bromine as a photosensitizer:

Bromine acts as a photosensitizer in the conversion of maleic acid into fumeric acid.



maleic acid fumeric acid.

d) cadmium vapour as photosensitizer:

Cadmium vapour acts as photosensitizer for the polymerization of ethylene.

$$n C_2H_4 \xrightarrow[h\nu]{Cd} (C_2H_4)_n.$$

e) Chlorophyll as a photosensitizer:

$$\text{Chlorophyll} + h\nu \longrightarrow [\text{Chlorophyll}]^*$$

$$6CO_2 + 6H_2O + [\text{Chlorophyll}]^* \xrightarrow{\text{sunlight}} C_6H_{12}O_6 + 6O_2 +$$

* Function of a photosensitizer: ↓ [Chlorophyll]

Absorb light → become excited → transfer energy to a reactant → activate them for reaction → without taking part in reaction.

2013

Explain the factors which affect fluorescence and phosphorescence.

Factors affecting fluorescence and phosphorescence:

(a) All the molecules cannot show the phenomena of fluorescence and phosphorescence. Only such molecules show these phenomena that are able to absorb ultraviolet or visible light. In general, the greater the absorbancy of a molecule, the more intense its luminescence.

b) substituents often exhibit a marked effect on the fluorescence and phosphorescence of molecules.

i) Electron withdrawing group like -COOH, $-NO_2$ etc. decrease or even destroy fluorescence.

ii) Electron releasing group like $-NH_2$ and -OH often enhance fluorescence.

iii) If a high atomic number atom is introduced into a π-electron system, it enhances phosphorescence and decrease fluorescence.

c) The pH exhibits a marked effect on the fluorescence of compounds. For example, alkaline soln of amine shows fluorescence in visible region but if the soln is acidified, the visible fluorescence disappears.

What is meant by photochemical equilibrium? Give example.

Photochemical equilibrium:

If the forward reaction and reversed reaction occurs both by light or one by light and another thermally, both are called photochemical equilibrium.

Two types:

1. First category: Only one reaction is light sensitive.

$$A + B \underset{\text{Dark}}{\overset{\text{light}}{\rightleftharpoons}} C + D$$

Example:

i. Dissociation of nitrogen di-oxide.

$$2NO_2 \underset{\text{Dark}}{\overset{\text{light}}{\rightleftharpoons}} 2NO + O_2$$

ii. Dimerisation of anthracene.

$$2\,C_{14}H_{10} \underset{\text{Dark}}{\overset{\text{light}}{\rightleftharpoons}} C_{28}H_{20}$$

2. second category: Both the reactions are light sensitive.

$$A + B \underset{\text{light}}{\overset{\text{light}}{\rightleftharpoons}} C + D$$

Example:

i. formation of sulphur tioxide.

$$2SO_2 + O_2 \rightleftharpoons 2SO_3.$$

*** short note

* What 2016, 2014 2

What is quantum yield? why does the value of quantum yield deviate from unity? Give examples where the quantum yield is greater than unity and less than unity.

Quantum yield:

Quantum yield is the number of molecules which ~~undeg~~ undergo chemical transformation per quantum of absorbed energy.

Mathematically,

$$\phi = \frac{\text{No. of molecules reacting in a given time}}{\text{No. of quanta absorbed in the same time}}$$

$$= \frac{\text{No. of moles reacting in a given time}}{\text{No. of einsteins absorbed in the same time.}}$$

$$= \frac{\text{Rate of chemical reaction}}{\text{No. of einsteins absorbed}}$$

There are two reasons for the deviation of quantum yield from unity –

i) If the number of molecules decomposes less than one per photon, the reaction has a low quantum yield, $\phi < 1$.

ii) when two or more molecules decomposed per photon, then the reaction has a high quantum yield, $\phi > 1$.

Examples of high quantum yield reactions are –

(i) The combination of carbon monoxide and chlorine in the presence of light.

$$CO + Cl_2 \xrightarrow{h\nu} COCl_2, \quad \phi = 10^3$$

(ii) The combination of hydrogen and chlorine to form hydrogen chloride by a wavelength less than 4500 A°

$$H_2 + Cl_2 \xrightarrow{h\nu} 2HCl, \quad \phi = 10^4 \sim 10^6.$$

Examples of low quantum yield reactions are—

(i) The combination of hydrogen and bromine to form hydrogen bromide.

$$H_2 + Br_2 \xrightarrow{h\nu} 2HBr, \quad \phi = 0.01.$$

(ii) Photolysis of ammonia by a wavelength of 2100 Å.

$$2NH_3 \xrightarrow{h\nu} N_2 + 3H_2, \quad \phi = \text{~~0.02~~} \; 0.2.$$

*** short note

Explain chemiluminescence with suitable examples.

2014

Chemiluminescence:

Chemiluminescence is the phenomenon in which the emission of visible light as a result of a chemical change at a temperature at which a black body normally does not emit visible light.

Example:

(i) When solution of strontium chloride is added to dilute H_2SO_4 in dark, a feeble glow is produced along with the precipitate of $SrSO_4$.

$$SrCl_2 + H_2SO_4 \longrightarrow SrSO_4 + 2HCl + h\nu.$$

(ii) The oxidation of phosphorous vapours by atmospheric oxygen produces a greenish-yellow glow is the classical example of chemiluminescence.

* How does photochemical reaction differ from thermal reaction? 2016

Difference between photochemical reaction and thermal reaction:

Photochemical reaction	Thermal reaction
1. Photochemical reaction initiated and influenced by light.	1. Thermal reactions initiated by collisions between reactant molecules and catalyst except light.
2. Reaction rate is independent on temperature. But the rate depends on the intensity of light.	2. Temperature has a significant effect on the rate of a thermochemical reaction.
3. ΔG for photochemical reaction may be positive or negative.	3. ΔG for thermochemical reaction is always negative.
4. Photochemical activation is highly selective in nature.	4. Thermochemical activation is not selective in nature.

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Photochemical reaction	Thermal reaction.
4. complese in nature.	5. simple in nature.
6. Example: $H_2 + Br_2 \xrightarrow{h\nu} 2HBr$	6. Example: $N_2 + 3H_2 \rightarrow 2NH_3$.

Fluorescence	Phosphorescence
1. The emission of light occurs at a short interval of time of the order 10^{-8} sec.	1. The emission of light occurs comparatively at longer time about one second or more
2. The energy of the fluorescence radiation will be less than that of the exciting radiation	2. The energy of the phosphorescence will be same as the exciting radiation
The process will take place relatively fast.	3. The process will take place relatively slow.
4. Example: Chlorophyll present green leaves show phenomenon of [illegible]	4. Example: sulphides of calcium, barium, strontium shows phosphorescence.

short note + question

~~** Discuss~~ about the primary photochemical process using molecular spectra. ✓ 2015

Primary photochemical process:

The process in which light radiation is absorbed by an atom or molecule giving rise to the formation of an excited atom or an excited molecule is called primary photochemical process.

$$A + h\nu \longrightarrow A^*$$

↑ Atom or molecule ↑ Quantum of energy ↖ Excited molecule or atom.

In the primary photochemical process, the absorption of energy raises the molecule from lower state to higher electronic state due to tramition of valance electrons in the molecule.

The excited molecules may then behave in different ways leading to different types of phenomena —

→ Bonding (σ and π)
→ Antibonding (σ^* and π^*)
→ Nonbonding (n)

~~Three kinds of molecular~~

The electronic transition consist of the excitation of ~~an~~ a electron from an occupied bonding or nonbonding molecular orbital to an antibonding molecular orbital.

i. $\sigma \rightarrow \sigma^*$ transition
ii. $\pi \rightarrow \pi^*$ transition
} → (i) and (ii) are strong transition at shorter wavelength

iii. $n \rightarrow \pi^*$ transition
iv. $n \rightarrow \sigma^*$ transition
} → (iii) and (iv) are weak transition at longer wavelength.

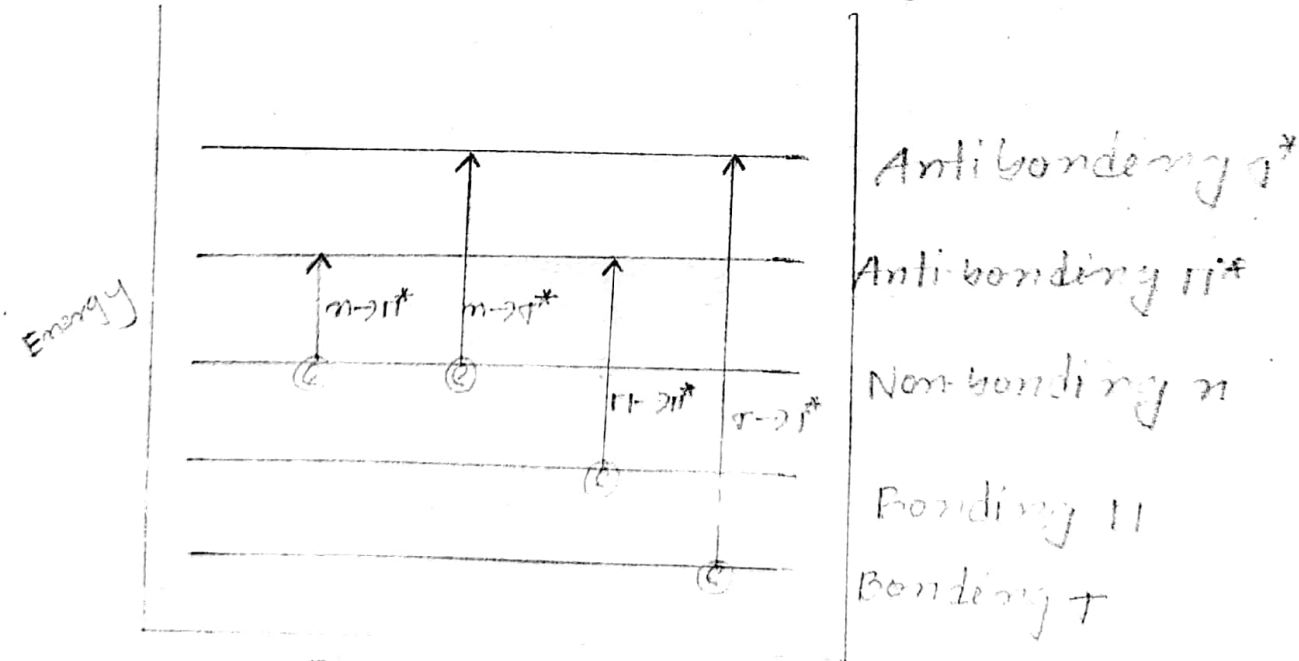


Fig: [illegible]

* 26. The dissociation energy of a molecule is 142.95 Kcal mol^{-1}. what is the longest wavelength of light capable of dissociation of molecule.

we know,

$$E = \frac{Nhc}{\lambda}$$

$$\Rightarrow \lambda = \frac{Nhc}{E}$$

where

$E = 142.95$ Kcal/mol

$= 142.95 \times 10^3 \times 4.184$ J/mol

$N = 6.023 \times 10^{23}$

$c = 3 \times 10^8$ ms^{-1}

$\lambda = ?$

$h = 6.626 \times 10^{-34}$ Js

$$= \frac{6.023 \times 10^{23} \times 6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{142.95 \times 10^3 \times 4.184 \text{ J/mol}}$$

$= 2001.75 \times 10^{-10}$ m

$= 2001.75$ Å

Ans:

* For a photochemical reaction 1.0×10^{-5} mol of a substance reacting to form a product on absorption of 6.62×10^{7} ergs at 3600 Å. calculate quantum yield of the reaction.

Here,

no. of molecules reacting $= 1.0 \times 10^{-5} \times 6.023 \times 10^{23}$

$= 6.023 \times 10^{18}$

Again, no. of quanta absorbed

$$= \frac{\text{Total energy absorbed}}{\text{Energy of one quantum}}$$

$$= \frac{6.62 \times 10^{7} \text{ erg}}{h\nu}$$

$$= \frac{6.62 \times 10^{7} \text{ erg} \times \lambda}{hc}$$

$$= \frac{6.62 \times 10^{7} \text{ erg} \times 3600 \times 10^{-10} \text{ m}}{6.626 \times 10^{-34} \text{ J s} \times 3 \times 10^{8} \text{ m s}^{-1}}$$

$= 1.2 \times 10^{19}$

$$\therefore \text{quantum yield} = \frac{\text{no. of molecules reacting}}{\text{no. of quanta absorbed}}$$

$$= \frac{6.023 \times 10^{23}}{1.2 \times 10^{19}}$$

$$= 0.502$$

Ans

* A substance absorbs 2.0×10^{16} quanta of radiation per second and 0.002 mol of it reacts in 120 seconds, calculate quantum yield of the reaction.

Here,

no. of molecules reacting

$$= 6.023 \times 10^{-23} \times 0.002$$

$$= 1.2046 \times 10^{21}$$

Again,

no. of quanta absorbed $= 2.0 \times 10^{16} \times 120$

$$= 2.908 \times 10^{19}$$

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Now, quantum yield = $\frac{\text{no. of molecules reacting}}{\text{no. of quanta absorbed}}$

$= \frac{1.2046 \times 10^{21}}{2.408 \times 10^{19}}$

$= 50.025$

Ans:

The dissociation energy of a molecule is 102.9 kcal mol^{-1}. If this or 102900 cal/mole or H_2 molecule is dissociated is dissociated by illumination with radiation of wavelength 2537 Å. what fraction of radiant energy will be converted into ~~K.E~~ kinetic energy?

Here, Dissociation energy per mole $= \frac{102.9 \times 10^3}{4.184}$ J

$= \frac{102.9 \times 10^3}{6.023 \times 10^{23}}$ J

$= 7.15 \times 10^{-19}$ J

Again,

Energy of one quantum $= \frac{hc}{\lambda}$

$= \frac{6.626 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{2537 \times 10^{-10} \text{ m}}$

$= 7.89 \times 10^{-19}$ J

∴ Energy converted into

$$K.E = (7.89 \times 10^{-19} - 7.15 \times 10^{-19})\ J$$

$$= 0.69 \times 10^{-19}\ J$$

∴ Fraction converted into $K.E = \frac{0.69 \times 10^{-19}}{7.84 \times 10^{-19}}$

$$= 0.088$$

Ans: